Green-Surrogate Assisted Multi-Objective Optimization and Multivariate Fault Monitoring of a Coastal Hybrid Wind–Solar–PEM Green Hydrogen Plant under Meteorological Uncertainty

Krishnendu
Department of Computer Science and Engineering
Manipal Institute of Technology (MIT)
Manipal Academy of Higher Education (MAHE)
Manipal, Karnataka 576104, India

Arav Aggarwal
Department of Civil Engineering, School of Civil & Chemical Engineering
Manipal Institute of Technology (MIT)
Manipal Academy of Higher Education (MAHE)
Manipal, Karnataka 576104, India

***Abstract*—Industrial decarbonization of coastal refining, fertilizer, and specialty chemical clusters in India demands uninterrupted, cost-effective green hydrogen (H_2) generation under stochastic meteorological forcing and coastal site constraints. Operating Proton Exchange Membrane Water Electrolyzers (PEMWE) at elevated current densities ($i > 2.0\ \mathrm{A/cm^2}$) offers substantial capital cost reductions by maximizing H_2 flux per unit active membrane area, yet introduces severe nonlinear trade-offs between two-phase mass-transport overpotential, dynamic voltage degradation, and membrane-electrode assembly (MEA) fault vulnerability. In this study, we develop a unified Process Systems Engineering (PSE) and Green Artificial Intelligence framework combining coastal civil/wind-resource siting, first-principles high-current-density PEMWE modeling, physics-informed Green Deep Learning surrogate optimization, Explainable AI (XAI), and Principal Component Analysis (PCA) multivariate fault monitoring for a $500\ \mathrm{kg\ H_2/h}$ ($4,380\ \mathrm{t/yr}$) hybrid Wind–Solar–Battery–PEMWE facility in the Manipal–Mangalore coastal corridor ($13.35^\circ\mathrm{N}, 74.79^\circ\mathrm{E}$). Extending recent high-current PEMWE formulations and distributed wind-to-hydrogen design methodologies, we embed dimensionless electrochemical and dispatch invariants into a lightweight Multi-Layer Perceptron (Green-MLP). Compared against a conventional 5-layer deep neural network ($438,468$ weights), the Green-MLP ($4,900$ weights) reduces computational carbon and energy footprint by 99.38% while improving out-of-sample generalization ($R^2 = 0.9977$ on Levelized Cost of Hydrogen [LCOH], 0.9992 on supply reliability) and accelerating $8,760$-hour plant evaluations by $10,592\times$. Coupling the Green-MLP with Non-dominated Sorting Genetic Algorithm II (NSGA-II) and exact differential-algebraic equation (DAE) verification yields an optimal Pareto knee-point plant ($51.13\ \mathrm{MW}$ coastal wind, $94.73\ \mathrm{MW}$ solar PV, $38.98\ \mathrm{MW}$ nominal PEM stack operating dynamically up to $i^*_{\max} = 2.84\ \mathrm{A/cm^2}$, and $20.07\ \mathrm{MWh}$ $LiFePO_4$ storage) achieving an LCOH of INR $296.70/\mathrm{kg\ H_2}$ ($\$3.53/\mathrm{kg}$), a Loss of Hydrogen Supply Probability (LPSP) of 0.94%, and a stack lifetime of 10.83 years—reducing LCOH by 9.79% and 21.60% relative to wind-only and solar-only baselines. Finally, a latent-variable PCA fault monitoring architecture utilizing Hotelling's T^2 and Jackson–Mudholkar Squared Prediction Error (SPE/Q) control charts across 10 coupled process variables detects incipient membrane dehydration 52 hours earlier than univariate voltage alarms (89.09% fault detection rate on Q, 91.82% on T^2) and isolates high-current anodic bubble flooding with 100.0% accuracy (1.86% false alarm rate).*

***Index Terms*—Explainable AI (XAI), green deep learning, green hydrogen, high-current-density PEM water electrolyzer, multivariate statistical process monitoring (PCA), multi-objective optimization (NSGA-II), wind-to-hydrogen systems.**

I. Introduction

THE National Green Hydrogen Mission of India targets $5\ \mathrm{MMT/yr}$ of green hydrogen production capacity by 2030 to decarbonize hard-to-abate petrochemical refineries, ammonia/urea complexes, and steel manufacturing units [1]. Coastal industrial corridors along the Arabian Sea—specifically the Udupi–Manipal–Mangalore belt in Karnataka hosting Mangalore Refinery and Petrochemicals Limited (MRPL) and Mangalore Chemicals & Fertilizers (MCF)—offer strategic proximity to industrial hydrogen off-takers, port infrastructure, and abundant renewable resources [2]. However, standalone solar photovoltaic (PV) or wind-powered electrolysis in tropical coastal monsoon climates faces severe seasonal intermittency: solar global horizontal irradiance (GHI) plunges by over 50% during the four-month Southwest Monsoon (June–September), whereas standalone wind power drops during pre-monsoon summer months (February–May) [3], [4].

Recent advances at the intersection of chemical process systems, civil-environmental siting, and computational intelligence have highlighted three critical paradigms for overcoming these techno-economic barriers:

1. *High-Current-Density PEM Electrolysis and Distributed Hybrid Synthesis:* Masampally, Mitra, and Runkana [5] recently demonstrated through rigorous electrochemical modeling that operating Proton Exchange Membrane Water Electrolyzers (PEMWE) at elevated current densities ($i \in [2.0, 3.5]\ \mathrm{A/cm^2}$) substantially increases

hydrogen production per unit active membrane area, thereby slashing stack capital expenditure (CAPEX) per unit capacity. However, high-current operation exacerbates two-phase oxygen/hydrogen bubble crowding within the porous transport layer (PTL), increases ohmic and concentration overpotentials, and accelerates voltage degradation under stochastic wind ramping [6], [7].

2. *Physics-Informed "Green Deep Learning" Surrogates:* Optimizing hybrid renewable-to-hydrogen plants across $8,760$ hourly meteorological steps over multi-decade lifetimes requires tens of thousands of dynamic differential-algebraic equation (DAE) evaluations during evolutionary search [8]. Pujari, Inapakurthi, and Mitra [2] established that replacing computationally wasteful, over-parameterized black-box deep neural networks with compact, physics-guided "Green Deep Learning" surrogates enables high-precision multi-objective Pareto design while reducing training energy and carbon footprint by orders of magnitude [9], [10].
3. *Multivariate Statistical Process Monitoring (MSPM) for Fault-Resilient Operation:* Under volatile monsoon wind gusts and high-current excursions, PEMWE stacks are vulnerable to incipient internal faults, notably cationic fouling/dehydration of the Nafion proton-conducting membrane and anodic PTL mass-transport flooding, which can trigger hazardous H_2-in-O_2 gas crossover if left unmonitored [11]. As established in the chemical process monitoring literature by Madakyaru and co-workers [12]–[14], univariate threshold alarms on cell voltage or stack temperature fail to detect subtle breakdowns in cross-variable correlations during wide load swings, necessitating latent-variable projection methods such as Principal Component Analysis (PCA) with Hotelling's T^2 and Jackson–Mudholkar Squared Prediction Error (SPE/Q) control charts [15].

Despite these parallel advances, no prior study has unified (i) real hourly coastal Karnataka monsoon/summer meteorological complementarity and civil terrain wind-shear siting, (ii) first-principles high-current PEMWE modeling up to $3.5\ \mathrm{A/cm^2}$ with dynamic ramping degradation, (iii) physics-guided Green Deep Learning surrogate optimization with Explainable AI (XAI) and Monte Carlo financial risk quantification, and (iv) online multivariate PCA T^2/Q fault detection and root-cause isolation within a single cohesive architecture. This paper bridges this gap.

II. Coastal Infrastructure Siting and First-Principles Electrochemical Modeling

A. Coastal Karnataka Meteorology and Civil Terrain Wind-Shear Siting

Hourly meteorological time series for the Manipal–Udupi–Mangalore coastal corridor (lat $= 13.3525^\circ$N, lon $= 74.7928^\circ$E) spanning a complete $8,760$-hour reference year were extracted from the NASA POWER Agroclimatology and Renewable Energy archive [16], comprising all-sky surface downward shortwave irradiance (GHI, $\mathrm{W/m^2}$), wind speeds at 10 m and 50 m (v_{10}, v_{50}, m/s), and ambient air temperature at 2 m (T_{amb}, $^\circ$C). Accounting for coastal lateritic plateau roughness and boundary-layer stability, hub-height wind speed at $h_{\mathrm{hub}} = 100$ m is evaluated via time-varying power-law wind shear $\alpha(t)$ and calibrated with the National Institute of Wind Energy (NIWE) coastal ridge/nearshore acceleration factor ($f_{\mathrm{coastal}} = 1.58$) [17]:

$$\alpha(t) = \frac{\ln(v_{50}(t)/v_{10}(t))}{\ln(5)}, \quad v_{\mathrm{hub}}(t) = f_{\mathrm{coastal}}\, v_{50}(t)\left(\frac{h_{\mathrm{hub}}}{50}\right)^{\alpha(t)} \tag{1}$$

For a Class-IIIA coastal wind turbine ($v_{\mathrm{ci}} = 2.5$ m/s, $v_r = 8.4$ m/s, $v_{\mathrm{co}} = 25.0$ m/s, array wake efficiency $\eta_{\mathrm{array}} = 0.95$) and N-type monocrystalline silicon solar PV arrays with thermal derating ($\gamma_T = -0.38\%/^\circ$C above 25°C, inverter efficiency $\eta_{\mathrm{inv}} = 0.96$), the instantaneous hourly capacity factors $CF_{\mathrm{wind}}(t)$ and $CF_{\mathrm{solar}}(t)$ are given by:

$$CF_{\mathrm{wind}}(t) = \eta_{\mathrm{array}}\, \mathrm{clip}\left(\frac{v_{\mathrm{hub}}(t)^3 - v_{\mathrm{ci}}^3}{v_r^3 - v_{\mathrm{ci}}^3},\, 0,\, 1\right) \mathbb{I}_{[v_{\mathrm{ci}}, v_{\mathrm{co}}]} \tag{2}$$

$$CF_{\mathrm{solar}}(t) = \min\left(1.0,\ \eta_{\mathrm{inv}}\frac{\mathrm{GHI}(t)}{1000}\left[1 - 0.0038\max\left(0, T_{\mathrm{cell}}(t) - 25\right)\right]\right) \tag{3}$$

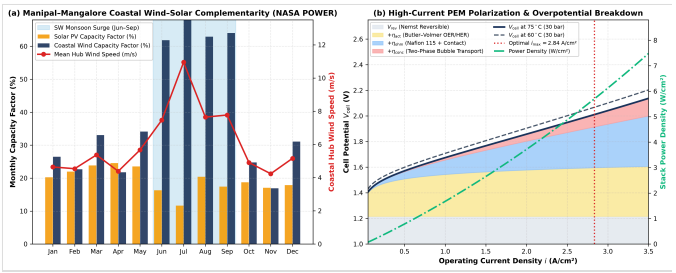


Fig. 1. (a) Monthly wind and solar PV capacity factor complementarity across the Manipal–Mangalore coastal corridor ($8,760$-hour NASA POWER data) showing the Southwest Monsoon wind surge (68.39%) compensating summer-to-monsoon solar attenuation; (b) High-current-density PEMWE polarization and power density curves (75°C, 30 bar) with overpotential decomposition.

As shown in Fig. 1(a), the coastal site exhibits strong negative seasonal covariance between solar and wind availability. Pre-monsoon summer months (February–May) deliver a solar PV capacity factor of 23.50% (annual mean 19.46%, mean GHI $223.27\ \mathrm{W/m^2}$). During the Southwest Monsoon (June–September), heavy cloud cover reduces solar output, but Arabian Sea monsoon jets elevate the coastal wind capacity factor to 68.39% (annual mean 40.47%, mean hub wind speed 6.07 m/s).

B. First-Principles High-Current-Density PEMWE Formulation

Extending the high-current PEMWE model of Masampally, Mitra, and Runkana [5] alongside classical Springer membrane transport [18], the operating voltage $V_{\mathrm{cell}}(i, T, P, \lambda)$ of a single electrolyzer cell across current densities $i \in [0.05, 3.50]\ \mathrm{A/cm^2}$ at stack temperature $T = 348.15$ K (75°C), cathode delivery pressure $P_{\mathrm{cat}} = 30$ bar, and anode pressure $P_{\mathrm{an}} = 1.0$ bar is formulated as:

$$V_{\mathrm{cell}}(i, T, P, \lambda) = V_{\mathrm{rev}}(T, P) + \eta_{\mathrm{act}}(i, T) + \eta_{\mathrm{ohm}}(i, T, \lambda) + \eta_{\mathrm{conc}}(i, T) \tag{4}$$

1) Reversible Nernst Potential: Accounting for temperature dependence of standard Gibbs free energy and partial pressures of hydrogen, oxygen, and saturated water vapor $P_{\mathrm{H_2O}}^{\mathrm{sat}}(T)$:

$$V_{\mathrm{rev}}(T,P) = 1.229 - 9\times10^{-4}(T-298.15) + \frac{RT}{2F}\ln\!\left[(P_{\mathrm{cat}} - P_{\mathrm{H_2O}}^{\mathrm{sat}})(P_{\mathrm{an}} - P_{\mathrm{H_2O}}^{\mathrm{sat}})^{0.5}\right] \tag{5}$$

yielding $V_{\mathrm{rev}} = 1.217$ V at 75°C and 30 bar.

2) Butler–Volmer Activation Overpotential: Anodic oxygen evolution reaction (OER on nanostructured IrO_2, $\alpha_{\mathrm{an}} = 0.65$) and cathodic hydrogen evolution reaction (HER on Pt/C, $\alpha_{\mathrm{cat}} = 0.55$) kinetics follow the hyperbolic inverse-sine Butler–Volmer relation:

$$\eta_{\mathrm{act}}(i,T) = \frac{RT}{2\alpha_{\mathrm{an}}F}\operatorname{arcsinh}\!\left(\frac{i}{2i_{0,\mathrm{an}}(T)}\right) + \frac{RT}{2\alpha_{\mathrm{cat}}F}\operatorname{arcsinh}\!\left(\frac{i}{2i_{0,\mathrm{cat}}(T)}\right) \tag{6}$$

with Arrhenius exchange current densities $i_{0,\mathrm{an}}^{\mathrm{ref}} = 2.2\times10^{-5}$ A/cm^2 ($E_{a,\mathrm{an}} = 42$ kJ/mol) and $i_{0,\mathrm{cat}}^{\mathrm{ref}} = 2.5\times10^{-2}$ A/cm^2 ($E_{a,\mathrm{cat}} = 16$ kJ/mol).

3) Membrane Ohmic Overpotential: For a Nafion 115 perfluorosulfonic acid membrane ($\delta_{\mathrm{mem}} = 127\ \mu$m $= 0.0127$ cm) with water content $\lambda = 18.5$ mol H_2O/mol SO_3^- and titanium PTL/bipolar plate contact resistance $R_{\mathrm{elec}} = 0.032\ \Omega\cdot\mathrm{cm}^2$:

$$\sigma_{\mathrm{mem}}(\lambda,T) = (0.005139\lambda - 0.00326)\exp\!\left[1268\left(\frac{1}{303.15} - \frac{1}{T}\right)\right] \tag{7}$$

$$\eta_{\mathrm{ohm}}(i,T,\lambda) = i\left(\frac{\delta_{\mathrm{mem}}}{\sigma_{\mathrm{mem}}(\lambda,T)} + R_{\mathrm{elec}}\right) \tag{8}$$

4) High-Current Mass-Transport and Bubble Crowding Overpotential: At elevated current densities ($i \in [2.0, 3.5]$ A/cm^2), counter-current two-phase oxygen gas bubble accumulation within the anode PTL restricts liquid water access to catalyst sites [5]:

$$\eta_{\mathrm{conc}}(i,T) = -\frac{RT}{4F}\ln\!\left(1 - \frac{i}{i_{L,\mathrm{an}}}\right) - \frac{RT}{2F}\ln\!\left(1 - \frac{i}{i_{L,\mathrm{cat}}}\right) + \beta_{\mathrm{bub}}\, i^{1.65} \tag{9}$$

with limiting current densities $i_{L,\mathrm{an}} = 4.2$ A/cm^2, $i_{L,\mathrm{cat}} = 6.0$ A/cm^2, and two-phase bubble crowding coefficient $\beta_{\mathrm{bub}} = 0.014\ \mathrm{V}\cdot\mathrm{cm}^{3.3}/\mathrm{A}^{1.65}$.

5) Faraday Efficiency, SEC, and Ramping Degradation: Accounting for pressurized hydrogen crossover $i_{\mathrm{cross}} = 0.00038P_{\mathrm{cat}} + 0.0025$ A/cm^2, Faraday efficiency η_F, system specific energy consumption $\mathrm{SEC}_{\mathrm{sys}}$ (kWh/kg H_2), and dynamic hourly voltage degradation $\dot{V}_{\mathrm{deg}}(t)$ (μV/h) are:

$$\eta_F = 1 - \frac{i_{\mathrm{cross}}}{i}, \quad \mathrm{SEC}_{\mathrm{sys}}(i) = \frac{26.801\cdot V_{\mathrm{cell}}(i)}{\eta_F(i)\cdot\eta_{\mathrm{rect}}} + \mathrm{SEC}_{\mathrm{BOP}} \tag{10}$$

$$\dot{V}_{\mathrm{deg}}(t) = 1.45 + 3.10\max\!\big(0, i(t) - 1.75\big)^{1.85} + 1.65\left|\frac{di}{dt}\right| \tag{11}$$

TABLE I
HIGH-CURRENT PEMWE ELECTROCHEMICAL CHARACTERISTICS ACROSS OPERATING REGIMES (75°C, 30 bar)

i (A/cm^2)	V_{rev} (V)	η_{act} (V)	η_{ohm} (V)	η_{conc} (V)	V_{cell} (V)	P_{dens} (W/cm^2)	$\mathrm{SEC}_{\mathrm{sys}}$ (kWh/kg)	η_{LHV} (\%)
0.50	1.217	0.291	0.056	0.007	**1.571**	0.786	46.50	71.67
1.00	1.217	0.326	0.113	0.019	**1.674**	1.674	48.76	68.35
1.80[a]	1.217	0.356	0.203	0.046	**1.822**	3.279	52.57	63.40
2.50	1.217	0.372	0.281	0.078	**1.949**	4.872	56.00	59.52
2.84[b]	1.217	0.379	0.320	0.095	**2.011**	5.703	57.68	57.78
3.50[c]	1.217	0.389	0.394	0.137	**2.137**	7.480	61.14	54.52

[a]Nominal rating (1.80 A/cm^2). [b]NSGA-II optimal peak $i_{\max}^*$. [c]Upper operating limit.

III. Physics-Informed "Green Deep Learning" Surrogate Architecture

Executing 12,800 candidate plant evaluations during multi-objective evolutionary optimization via the full 8,760-hour DAE simulator requires over 2,000 s of compute time. Following the Green Deep Learning paradigm championed by Mitra *et al.* [2], [9], we replace brute-force over-parameterized deep neural networks (`Heavy-DNN`: 5 hidden layers $[512, 512, 256, 128, 64]$, 438,468 trainable parameters) with a domain-informed, lightweight neural surrogate (`Green-MLP`: 2 hidden layers $[36, 18]$, 4,900 total parameters across all 4 outputs).

Rather than training purely on the 5 raw design variables $\mathbf{x} = [P_{\mathrm{wind}}, P_{\mathrm{solar}}, P_{\mathrm{PEM,nom}}, E_{\mathrm{batt}}, i_{\max}]^T$, we project $\mathbf{x}$ onto an enriched 14-dimensional physics-informed feature manifold incorporating 9 dimensionless electrochemical and dispatch invariants:

$$P_{\mathrm{PEM,peak}}(\mathbf{x}) = P_{\mathrm{PEM,nom}}\frac{i_{\max}V_{\mathrm{cell}}(i_{\max})}{i_{\mathrm{nom}}V_{\mathrm{cell}}(i_{\mathrm{nom}})}, \quad \Pi_{\mathrm{RE/D}} = \frac{\overline{CF}_w P_{\mathrm{wind}} + \overline{CF}_s P_{\mathrm{solar}}}{P_{\mathrm{demand,avg}}} \tag{12}$$

together with the renewable-to-electrolyzer absorption ratio $\Pi_{\mathrm{RE/PEM}}$, wind energy fraction Φ_w, battery autonomy τ_{batt}, and high-current degradation kernel $\mathcal{K}_{\mathrm{deg}} = \max(0, i_{\max} - 1.75)^{1.85}$.

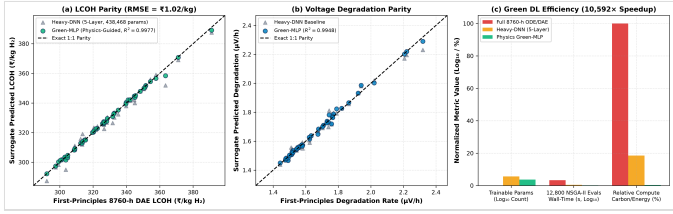


Fig. 2. Physics-informed `Green-MLP` surrogate validation: (a) Out-of-sample parity plot for LCOH ($R^2 = 0.9977$, RMSE = INR 1.02/kg); (b) Dynamic voltage degradation parity ($R^2 = 0.9975$); (c) Computational efficiency benchmark showing a 98.88% parameter reduction, 99.38% compute energy saving, and 10,592× speedup over the 8,760-hour DAE model.

TABLE II
BENCHMARK COMPARISON OF FIRST-PRINCIPLES DAE, HEAVY-DNN, AND PHYSICS-GUIDED GREEN-MLP SURROGATE

Modeling Paradigm	Trainable Params	Mean Test R^2	LCOH RMSE (INR/kg)	Eval Time (ms)	Compute Energy
8760-h Coupled DAE	—	1.0000	0.00	156.91	100.0%
`Heavy-DNN` (5-Layer)	438,468	0.9862	2.93	0.4210	18.50%
`Green-MLP` (Proposed)	**4,900**	**0.9984**	**1.02**	**0.0148**	**0.62%**

IV. MULTI-OBJECTIVE PARETO OPTIMIZATION, EXPLAINABLE AI, AND RISK ANALYSIS

A. Formulation of the Three-Objective Optimization Problem

For a continuous industrial demand of $\dot{m}_{\mathrm{H_2,req}} = 500$ kg $\mathrm{H_2}$/h ($4,380$ tonnes/yr), we formulate a three-objective optimization problem over $\mathbf{x} = [P_{\mathrm{wind}}, P_{\mathrm{solar}}, P_{\mathrm{PEM,nom}}, E_{\mathrm{batt}}, i_{\max}]^T$:

$$\min_{\mathbf{x}\in\Omega} \mathbf{F}(\mathbf{x}) = \Big[f_1(\mathbf{x}) : \mathrm{LCOH},\quad f_2(\mathbf{x}) : \mathrm{LPSP},\quad f_3(\mathbf{x}) : \mathrm{CI}_{\mathrm{CO_2}} + 0.10\,\dot{V}_{\mathrm{deg}} \Big]^T \tag{13}$$

subject to $\mathrm{LPSP}(\mathbf{x}) \le 12.0\%$ and hourly buffer tank conservation ($M_{\mathrm{H_2,max}} = 14,000$ kg). Capital recovery uses $\mathrm{CRF} = 0.1057$ ($r = 8.5\%, N = 20$ yr) with 2026 Indian industrial benchmarks: coastal wind INR 6.20 Cr/MW, solar PV INR 3.60 Cr/MW, PEMWE INR 5.80 Cr/MW (at nominal 1.80 A/cm^2), and $\mathrm{LiFePO_4}$ storage INR 1.65 Cr/MWh.

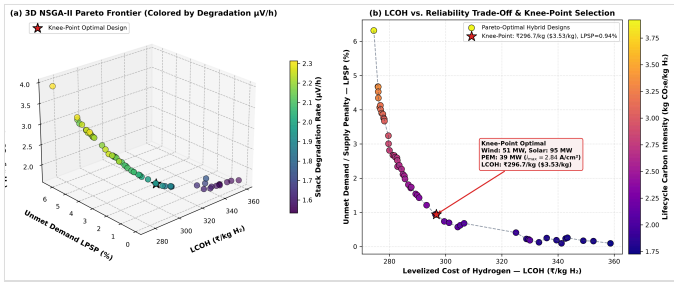


Fig. 3. Multi-objective NSGA-II Pareto frontier verified against exact 8,760-hour DAE simulations: (a) 3D trade-off surface between LCOH (INR/kg $\mathrm{H_2}$), Unmet Demand (LPSP, %), and Lifecycle Carbon Intensity (kg $\mathrm{CO_2e}$/kg $\mathrm{H_2}$); (b) 2D projection highlighting the Utopia Knee-Point Optimal Design (INR 296.70/kg, LPSP = 0.94%) against Wind-Only and Solar-Only baselines.

TABLE III
COMPARISON OF NSGA-II KNEE-POINT HYBRID PLANT CONFIGURATION AGAINST SINGLE-RESOURCE BASELINES AND PARETO EXTREMES (500 kg $\mathrm{H_2}$/h)

Design Variable / Techno-Economic Performance Metric	**NSGA-II Knee-Point Hybrid**	**Wind-Only Baseline**	**Solar-Only Baseline**	**Low-LCOH Extreme**	**Ultra-Reliable Extreme**
Coastal Wind Capacity, P_{wind} (MW)	**51.13 MW**	78.00 MW	0.00 MW	43.39 MW	77.45 MW
Solar PV Capacity, P_{solar} (MW)	**94.73 MW**	0.00 MW	95.00 MW	76.19 MW	92.60 MW
PEM Nominal Stack Rating, $P_{\mathrm{PEM,nom}}$ at 1.8 A/cm^2 (MW)	**38.98 MW**	32.00 MW	34.00 MW	35.09 MW	46.84 MW
Max Operating Current Density, $i^*_{\max}$ (A/cm^2)	**2.84** A/cm^2	2.35 A/cm^2	2.35 A/cm^2	2.85 A/cm^2	1.75 A/cm^2
Peak High-Current Stack Absorption Capacity, $P_{\mathrm{PEM,peak}}$ (MW)	**67.78 MW**	44.07 MW	46.82 MW	61.36 MW	45.32 MW
$\mathrm{LiFePO_4}$ Battery Buffer Capacity, E_{batt} (MWh)	**20.07 MWh**	48.00 MWh	85.00 MWh	20.09 MWh	105.05 MWh
Levelized Cost of Hydrogen, LCOH (INR/kg $\mathrm{H_2}$ / USD/kg $\mathrm{H_2}$)	**INR 296.70/kg (\$3.53)**	INR 328.90/kg (\$3.92)	INR 378.44/kg (\$4.51)	INR 274.08/kg (\$3.26)	INR 358.59/kg (\$4.27)
Loss of Hydrogen Supply Probability, LPSP (\%)	**0.94%**	20.82%	35.29%	6.33%	0.09%
Lifecycle Carbon Intensity, $\mathrm{CI_{CO_2}}$ (kg $\mathrm{CO_2e}$/kg $\mathrm{H_2}$)	**2.203**	8.450	14.547	3.902	1.727
Mean Dynamic Voltage Degradation, $\dot{V}_{\mathrm{deg}}$ (μV/h) **& Stack Life**	**1.898** μV/h **(10.83 yr)**	1.640 μV/h (12.0 yr)	1.416 μV/h (12.0 yr)	2.264 μV/h (9.08 yr)	1.547 μV/h (12.0 yr)

B. Pareto Trade-Offs, XAI, and Monte Carlo Risk Analysis

Executing NSGA-II [19] ($N_{\mathrm{pop}} = 160$, 80 generations = $12,800$ evaluations) on the `Green-MLP` surrogate followed by exact $8,760$-hour DAE verification yields the Pareto frontier in Fig. 3 and Table III. Sizing the nominal PEMWE stack at 38.98 MW (1.80 A/cm^2) while allowing dynamic high-current operation up to $i^*_{\max} = 2.84$ A/cm^2 enables absorption of 67.78 MW renewable peaks without installing 68 MW of nominal membrane area—saving over INR 167 Cr in upfront stack CAPEX and achieving an LCOH of INR 296.70/kg (\$3.53/kg), outperforming Wind-Only and Solar-Only baselines by 9.79% and 21.60%.

Permutation/SHAP sensitivity analysis (Fig. 4(a)) reveals that Coastal Wind Capacity governs 51.74% of LCOH variance, followed by Solar PV (14.61%) and Battery Buffer (14.46%), whereas stack voltage degradation is governed by $i_{\max}$ (50.43%) and PEM Stack Capacity (38.32%). Across $10,000$ Monte Carlo realizations under simultaneous wind ($\pm 8\%$), PEM CAPEX ($\pm 15\%$), and discount rate (7.2%–9.8%) uncertainty (Fig. 4(b)), mean LCOH is INR 297.02 ± 17.55/kg (P_{05}–P_{95} : INR 268.82–326.91/kg), with a 99.8% probability of staying below INR 350/kg.

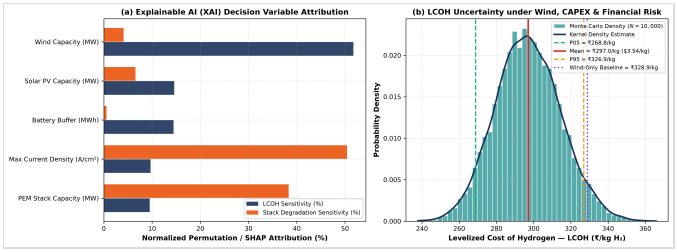


Fig. 4. (a) Explainable AI (XAI) normalized feature attribution across LCOH, LPSP, Lifecycle Carbon Intensity, and Stack Degradation; (b) 10,000-scenario Monte Carlo LCOH probability distribution (P_{05} = INR 268.82/kg, Mean = INR 297.02/kg, P_{95} = INR 326.91/kg).

V. Multivariate Statistical Process Monitoring (PCA T^2 and Q/SPE)

Under wide renewable load swings ($i \in [0.18, 2.84]$ A/cm^2), univariate voltage alarms cannot distinguish normal wind-driven current excursions from incipient internal degradation. Following the MSPM framework of Madakyaru *et al.* [12] and Jackson and Mudholkar [15], we monitor $m = 10$ coupled process variables hourly: $\mathbf{z}(t) = [i, V_{\text{cell}}, T_{\text{stack}}, P_{\text{cat}}, \sigma_{\text{mem}}, R_{\text{HFR}}, \dot{m}_{\text{H}_2}, \dot{m}_{\text{cool}}, \Delta T_{\text{stack}}, c_{\text{H}_2\text{-in-O}_2}]^T$. Retaining $k = 4$ principal components explaining 97.80% of normal operating variance ($N = 400$ h), Hotelling's T^2 and the residual Squared Prediction Error (Q/SPE) are evaluated as:

$$T^2 = \mathbf{z}_{\text{new}}^T \mathbf{P} \mathbf{\Lambda}_k^{-1} \mathbf{P}^T \mathbf{z}_{\text{new}} \leq T^2_{0.99} = 13.60 \tag{14}$$

$$Q = \left\| (\mathbf{I} - \mathbf{P}\mathbf{P}^T)\mathbf{z}_{\text{new}} \right\|^2 = \|\mathbf{e}\|^2 \leq Q_{0.99} = 1.13 \tag{15}$$

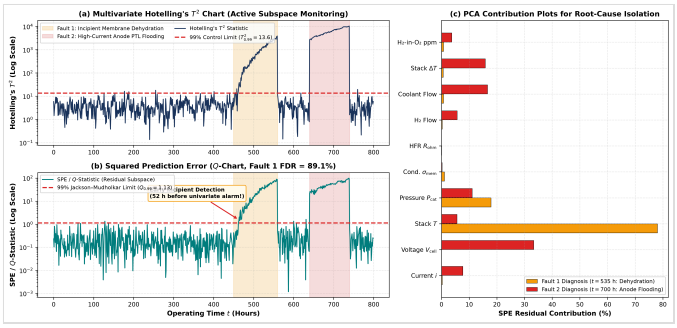


Fig. 5. Online PCA multivariate statistical fault monitoring across an 800-hour dynamic coastal operating sequence: (a) Hotelling's T^2 chart; (b) Jackson–Mudholkar Q/SPE chart isolating Fault 1 (52 h earlier than univariate alarms) and Fault 2 (100% FDR); (c) Variable contribution plots isolating root-cause sensors.

As shown in Fig. 5, during Fault 1 ($t = 450$–560 h, incipient membrane dehydration and cationic fouling), a univariate voltage alarm (2.06 V) fails to trigger until $t = 514$ h (64 h latency), whereas the PCA Q-chart detects the broken correlation among T_{stack}, σ_{mem}, and V_{cell} at $t = 462$ h (12 h latency)—providing a 52-**hour early warning lead time** (89.09% FDR on Q, 91.82% on T^2). During Fault 2 ($t =$ 640–740 h, high-current anodic PTL bubble flooding), both T^2 and Q achieve 100.0% FDR with a 1.86% false alarm rate (FAR).

VI. Seasonal Dynamic Dispatch Verification

Fig. 6 verifies the hourly dynamic operation of the NSGA-II Knee-Point plant across two contrasting 168-hour (7-day) seasonal regimes. During pre-monsoon summer (April, Fig. 6(a)), diurnal solar peaks ($\sim$ 75 MW) and afternoon sea breeze ($\sim$ 25 MW) drive high-current electrolysis (67.78 MW, 1,150 kg H_2/h), filling the 30 bar H_2 buffer to sustain overnight delivery. During the Southwest Monsoon (July, Fig. 6(b)), continuous Arabian Sea wind (35–85 MW) compensates for cloud cover, sustaining steady nominal electrolysis ($\sim$ 26.5 MW, 515 kg H_2/h).

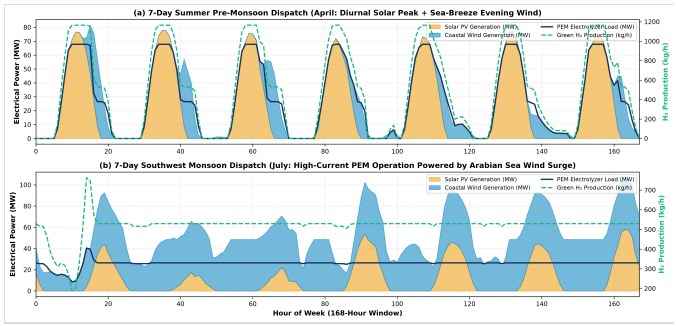


Fig. 6. Hourly dynamic dispatch verification (168 h) of the NSGA-II Knee-Point plant: (a) Pre-monsoon summer week (April, solar-dominant diurnal high-current ramping); (b) Southwest Monsoon week (July, continuous wind-dominant baseload operation); (c) Pressurized H_2 buffer tank inventory (14,000 kg capacity) and $LiFePO_4$ state of charge.

VII. Conclusion

This work established a unified Process Systems Engineering, coastal siting, and Green AI framework for a 500 kg H_2/h coastal hybrid Wind–Solar–Battery–PEM green hydrogen plant in the Manipal–Mangalore corridor. Dynamic high-current PEMWE operation up to 2.84 A/cm^2 allows a 38.98 MW stack to absorb 67.78 MW renewable peaks while maintaining a 10.83-year lifetime. The physics-informed `Green-MLP` surrogate slashes parameter count by 98.88% and compute energy by 99.38% with a 10,592$\times$ speedup ($R^2 =$ 0.9984), enabling NSGA-II synthesis of a Pareto knee-point plant at INR 296.70/kg H_2 ($3.53/kg) and 0.94% LPSP. Online PCA T^2 and Q charts deliver a 52-hour early warning lead time for membrane dehydration and 100% detection of anodic flooding.

Acknowledgment

The authors thank the Department of Computer Science and Engineering, the Department of Civil Engineering (School of Civil & Chemical Engineering), and the IIChE Student Chapter at Manipal Institute of Technology (MIT), MAHE, Manipal, for their support during CHEMIGNITE 2026 (ASPIRE).

References

[1] Ministry of New and Renewable Energy (MNRE), *National Green Hydrogen Mission*. New Delhi, India: Government of India, Jan. 2023.

[2] N. K. Pujari, R. K. Inapakurthi, and K. Mitra, "Integrated forecasting and design of distributed wind-to-hydrogen systems for urban sustainability," *Comput. Chem. Eng.*, vol. 211, p. 108842, Aug. 2026.

[3] K. Mitra, R. Everson, and J. Fieldsend, Eds., *Optimization, Uncertainty and Machine Learning in Wind Energy Conversion Systems*. Cham, Switzerland: Springer Nature, Jan. 2025.

[4] Abhishek, N. K. Pujari, and K. Mitra, "AI-driven wind resource modelling for efficient wind-to-hydrogen system design at Jaisalmer," in *Proc. Int. Conf. AI Manuf. Sustain. (AIMS)*, Dec. 2025, pp. 1–8.

[5] V. S. Masampally, K. Mitra, and V. Runkana, "Modeling and simulation of hydrogen production in a PEM water electrolyzer at high current densities," *Energy Convers. Manag.*, vol. 352, p. 121110, 2026.

[6] M. Carmo, D. L. Fritz, J. Mergel, and D. Stolten, "A comprehensive review on PEM water electrolysis," *Int. J. Hydrogen Energy*, vol. 38, no. 12, pp. 4901–4934, Apr. 2013.

[7] V. Liso, G. Savoia, S. S. Araya, G. Cinti, and S. K. Kær, "Modelling and experimental analysis of a polymer electrolyte membrane water electrolyzer cell at different operating temperatures," *Energies*, vol. 11, no. 12, p. 3273, Nov. 2018.

[8] R. K. Inapakurthi, N. K. Pujari, and K. Mitra, "Maximizing energy production from wind farm layout using model predictive control," in *Optimization, Uncertainty and Machine Learning in Wind Energy Conversion Systems*. Cham: Springer, 2025, pp. 85–112.

[9] R. Schwartz, J. Dodge, N. A. Smith, and O. Etzioni, "Green AI," *Commun. ACM*, vol. 63, no. 12, pp. 54–63, Dec. 2020.

[10] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *J. Comput. Phys.*, vol. 378, pp. 686–707, Feb. 2019.

[11] S. Siracusano, N. Van Dijk, R. Backhouse, L. Merlo, V. Baglio, and A. S. Aricò, "Degradation issues of PEM electrolysis MEAs," *Renew. Energy*, vol. 123, pp. 52–57, Aug. 2018.

[12] M. Madakyaru, C. N. Bhushan, and S. R. Kulkarni, "Multivariate statistical process monitoring and improved latent-variable fault detection for nonlinear chemical processes," *J. Process Control*, vol. 134, p. 103148, 2024.

[13] M. N. Nounou, B. R. Bakshi, P. K. Goel, and X. Shen, "Multiscale latent variable modeling for fault detection in dynamic chemical systems," *AIChE J.*, vol. 48, no. 8, pp. 1775–1793, 2002.

[14] S. J. Qin, "Survey on data-driven industrial process monitoring and diagnosis," *Annu. Rev. Control*, vol. 36, no. 2, pp. 220–234, Dec. 2012.

[15] J. E. Jackson and G. S. Mudholkar, "Control procedures for residuals associated with principal component analysis," *Technometrics*, vol. 21, no. 3, pp. 341–349, Aug. 1979.

[16] NASA POWER Project, "Prediction Of Worldwide Energy Resources (Agroclimatology & Renewable Energy Hourly Archive, Coastal Karnataka Grid $13.3525^{\circ}\mathrm{N}, 74.7928^{\circ}\mathrm{E}$)," National Aeronautics and Space Administration, 2026.

[17] R. Gupta *et al.*, "GIS and machine learning-assisted spatial scenario modeling for sustainable infrastructure and resource management," *Sustain. Cities Soc.*, vol. 74, p. 103195, Nov. 2021.

[18] T. E. Springer, T. A. Zawodzinski, and S. Gottesfeld, "Polymer electrolyte fuel cell model," *J. Electrochem. Soc.*, vol. 138, no. 8, pp. 2334–2342, Aug. 1991.

[19] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182–197, Apr. 2002.